SIX CLOSURE PROPERTIES OF THE QUASI-COHERENT MODULES ON A BIG SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Part (6) of the proof contains the author's parenthetical '(details omitted)' attached to the assertion that the problem is Zariski local on S; the reduction is routine and the remainder of the argument is displayed. The parenthetical is reproduced verbatim and nothing was supplied in its place.

Editorial note. The counted spans carry 1 navigation stub; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The equivalence between quasi-coherent modules on the big site of S and quasi-coherent modules on S, and the limits it preserves. Same chapter, Lemma lemma-equivalence-quasi-coherent-limits, invoked at proof lines 3544-3545 and then repeatedly, for instance at proof lines 3551-3552 and 3559-3560. Identifies each quasi-coherent module on the big site with the pullback of a quasi-coherent module on S and states which colimits and kernels that identification preserves.
- The corresponding closure facts for quasi-coherent modules on a scheme. Schemes, Section schemes-section-quasi-coherent, invoked at proof lines 3542-3543. Supplies the six statements on S that are being transferred.
- The behaviour of finite locally free modules under the pullback to the big site. Same chapter, Lemma lemma-equivalence-quasi-coherent-properties, invoked at proof lines 3590-3591. Shows that a quasi-coherent module on the big site whose pullback is finite locally free comes from a finite locally free module on S, which is what reduces part (6) to the finite free case.

Source reference. Original lemma 06VE: sources/stacks-descent/pinned-original-source.tex, source lines 3256–3256. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 05VG: sources/stacks-descent/pinned-original-source.tex, source lines 2830–2830. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original section 01LA: sources/stacks-chow/tags.txt, source lines 2079–2079. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let $\tau \in \{Zariski, étale, smooth, syntomic, fppf\}$. The category $QCoh((Sch/S)_\tau, \mathcal{O})$ of quasi-coherent modules on $(Sch/S)_\tau$ has the following properties:*

1. *Any direct sum of quasi-coherent sheaves is quasi-coherent.*
2. *Any colimit of quasi-coherent sheaves is quasi-coherent.*
3. *The cokernel of a morphism of quasi-coherent sheaves is quasi-coherent.*
4. *Given a short exact sequence of $\mathcal{O}$-modules $0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$ if $\mathcal{F}_1$ and $\mathcal{F}_3$ are quasi-coherent so is $\mathcal{F}_2$.*
5. *Given two quasi-coherent $\mathcal{O}$-modules the tensor product is quasi-coherent.*
6. *Given two quasi-coherent $\mathcal{O}$-modules $\mathcal{F}$, $\mathcal{G}$ such that $\mathcal{F}$ is finite locally free, the internal hom $\mathcal{H}om_{\mathcal{O}}(\mathcal{F}, \mathcal{G})$ is quasi-coherent.*

Proof. The corresponding facts hold for quasi-coherent modules on the scheme S, see Schemes, Section 01LA. The proof will be to use Lemma 06VE to transfer these truths to $(Sch/S)_\tau$.

Proof of (1). Let $\mathcal{F}_i$, $i \in I$ be a family of objects of $QCoh((Sch/S)_\tau, \mathcal{O})$. Write $\mathcal{F}_i = \mathcal{G}_i^a$ for some quasi-coherent modules $\mathcal{G}_i$ on S. Then $\bigoplus \mathcal{F}_i = (\bigoplus \mathcal{G}_i)^a$ by the lemma cited and we conclude.

Proof of (2). Let $\mathcal{I} \to QCoh((Sch/S)_\tau, \mathcal{O})$, $i \mapsto \mathcal{F}_i$ be a diagram. Write $\mathcal{F}_i = \mathcal{G}_i^a$ so we get a diagram $\mathcal{I} \to QCoh(\mathcal{O}_S)$. Then $\operatorname{colim} \mathcal{F}_i = (\operatorname{colim} \mathcal{G}_i)^a$ by the lemma cited and we conclude.

Proof of (3). Let $a : \mathcal{F} \to \mathcal{F}'$ be an arrow of $QCoh((Sch/S)_\tau, \mathcal{O})$. Write $a = b^a$ for some map $b : \mathcal{G} \to \mathcal{G}'$ of quasi-coherent modules on S. By the lemma cited we have $\operatorname{Coker}(a) = \operatorname{Coker}(b)^a$ (because a cokernel is a colimit) and we conclude.

Proof of (4). Write $\mathcal{F}_1 = \mathcal{G}_1^a$ and $\mathcal{F}_3 = \mathcal{G}_3^a$ with $\mathcal{G}_i$ quasi-coherent on S. By Lemma 06VE part (10) we conclude.

Proof of (5). Let $\mathcal{F}$ and $\mathcal{F}'$ be in $QCoh((Sch/S)_\tau, \mathcal{O})$. Write $\mathcal{F} = \mathcal{G}^a$ and $\mathcal{F}' = (\mathcal{G}')^a$ with $\mathcal{G}$ and $\mathcal{G}'$ quasi-coherent on S. By the lemma cited we have $\mathcal{F} \otimes_{\mathcal{O}} \mathcal{F}' = (\mathcal{G} \otimes_{\mathcal{O}_S} \mathcal{G}')^a$ and we conclude.

Proof of (6). Write $\mathcal{F} = \mathcal{H}^a$ for some quasi-coherent $\mathcal{O}_S$-module. By Lemma 05VG we see that $\mathcal{H}$ is finite locally free. The problem is Zariski local on S (details omitted) hence we may assume $\mathcal{H} = \mathcal{O}_S^{\oplus n}$ is finite free. Then $\mathcal{F} = \mathcal{O}^{\oplus n}$ and $\mathcal{H}om_{\mathcal{O}}(\mathcal{F}, \mathcal{G}) = \mathcal{G}^{\oplus n}$ is quasi-coherent. □